AI and the Distribution of Prosperity:

A Unified Framework for Understanding How Technology Shapes Inequality

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Abstract

The dividing line between AI utopia and dystopia is distribution. This paper develops a unified framework for understanding how AI affects the personal income distribution: whether its gains will be broadly shared or captured by a few. We begin by decomposing income inequality into three components: labor-income concentration, capital-income concentration, and the capital-labor split. We then trace the mechanisms through which AI affects these components in three steps: (i) *direct effects on production*, including automation, augmentation, new-task creation, changes in entry barriers, and capital deepening; (ii) the *division of productivity gains* between AI suppliers, adopting firms, workers, and consumers, as well as within each group; and (iii) *economy-wide demand effects* generated by the spending and saving of AI's beneficiaries. For each mechanism, we identify the key factors that determine its direction and magnitude, and synthesize the evidence to date. Finally, we discuss how these mechanisms can combine to produce concentrated gains or shared prosperity, and identify empirical indicators that can signal which direction the economy is heading.

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"But what of the future? Suppose that automation should drastically reduce [the labor share ...]. There would be a limited number of exceedingly wealthy property owners; the proportion of the working population required to man the extremely profitable automated industries would be small; wage-rates would thus be depressed; there would have to be a large expansion of the production of the labour-intensive goods and services which were in high demand by the few multi-multi-multi-millionaires; we would be back in a super-world of an immiserised proletariat and of butlers, footmen, kitchen maids, and other hangers-on." (Meade, 1964)

1 Introduction

Artificial intelligence can affect the economy in profoundly different ways. At one extreme, it could raise productivity across the economy, make essential services broadly accessible, and rebuild the middle class (Baily et al., 2023; Autor, 2024). At the other extreme, it could concentrate enormous gains among a handful of superstar firms, wealthy asset owners, and highly paid workers, while leaving the majority of workers behind (Acemoglu, 2025a). Where the economy ultimately falls between these two extremes will have profound economic, social, and political consequences. This is ultimately a question of *distribution*.

In this paper, we develop a framework for thinking through AI's effects on the entire *personal* income distribution and not just the factor income distribution. We focus on the distribution of total income including capital income and not just that of wages.[1] Understanding AI's distributional consequences requires identifying the mechanisms at play. Yet existing discussions are often fragmented: different mechanisms—such as machines replacing workers, AI raising worker productivity, firms earning higher profits, goods becoming cheaper, and gains flowing to those who own the technology—are usually dis-

[1] Atkinson and Lakner (2021) and Gomez (2025) similarly decompose the income share of the top group into its labor and capital components. Dalton (1920) emphasized the distinction between personal and factor income distribution already more than one hundred years ago. Describing his budding interest in studying the income distribution, Dalton (1920) cited in Atkinson (1997), writes: "I gradually noticed, however, that most 'theories of distribution' were almost wholly concerned with distribution as between 'factors of production'. Distribution as between persons, a problem of more direct and obvious interest, was either left out of the textbooks altogether, or treated so briefly, as to suggest that it raised no question, which could not be answered either by generalizations about the factors of production, or by plodding statistical investigations, which professors of economic theory were content to leave to lesser men."

cussed in isolation.[2] At the same time, the sheer number of possible forces can make the question seem overwhelmingly complex.

We begin in Section 2 by decomposing income inequality into three components: the concentration of labor income, the concentration of capital income, and the division of national income between labor and capital. Throughout this paper, we focus on the *pre-distribution* (i.e. pre-tax-and-transfer distribution) of income. We do so for two main reasons. First, technology shapes the income distribution primarily through markets—wages, profits, and prices—so its direct effects are reflected mostly in pre-distribution incomes. Taxes and transfers are the policy response to those effects, and informing that response is one of this paper's objectives. Second, as Appendix A.1 shows, post-tax inequality has historically tracked pre-tax inequality closely, both across countries and within them over time. In the U.S., redistribution offset only a small fraction of the post-1980 rise in top income shares (Piketty et al., 2018). We establish where the concentration of labor income, capital income, and the division between the two currently stand as AI diffuses broadly through the economy: wage inequality is high by historical standards, capital income is very concentrated after decades of rising wealth inequality, and the labor share has declined substantially over the past two decades. We close the section by distinguishing changes in inequality from changes in income *levels*: with enough growth, incomes can rise even at the bottom of the distribution while inequality increases. By focusing on the *distribution* of income, the paper encompasses both.

Section 3 then organizes the main mechanisms through which AI can affect these components into three closely interconnected categories. The first category (Section 3.1) consists of *direct effects on production*: automation, augmentation, new-task creation, changes in entry barriers, and capital deepening, closely paralleling the five categories of technological change proposed by Acemoglu et al. (2026). The distributional effects of these mechanisms depend primarily on AI's capabilities and on how workers and firms adopt and use it. The second category (Section 3.2) concerns *the distribution of the productivity*

[2]Existing distributional analyses often only focus on either the factor income distribution, i.e. the aggregate capital-labor split (like in the labor share literature), or the wage distribution. See, for example, Acemoglu and Restrepo (2019a); Berg et al. (2018); Korinek and Stiglitz (2019); Autor and Kausik (2026); Korinek et al. (2026) for analyses of the factor income distribution. Such theories—for example, "two class models" with capitalists and workers—cannot speak to a number of empirical regularities in developed countries, for example, that individuals at the top of the labor income distribution typically also earn substantial capital incomes (and vice versa). See, for example, Webb (2019); Freund and Mann (2026); Althoff and Reichardt (2026); Hosseini and Lichtinger (2026a) for analyses of AI's effects on the wage distribution.

gains generated by AI adoption. This distribution occurs through two splits: (i) the division of gains between AI suppliers and adopting firms through the prices charged for AI inputs; and (ii) the division of the gains remaining at adopting firms among owners, workers, and consumers—as well as within each category—through higher wages, lower prices, and higher profits.[3] The first split depends largely on the degree of competition and bottlenecks along the AI supply chain; the second depends on adopting firms' product-market power, workers' bargaining power, and the responsiveness of consumer demand to lower prices. Finally, the third category (Section 3.3) consists of *economy-wide demand effects*: as the initial beneficiaries of AI spend or save their gains, they reshape demand for different goods and services and, in turn, for the workers who produce them. These effects depend on who directly receives AI's gains, how much they spend or save, and which goods and services they spend on.

Together, these three categories trace how AI's economic effects unfold: how it changes production, how the resulting productivity gains are initially distributed, and how those gains subsequently reshape demand throughout the economy. At each category, we identify the key technological and economic factors that determine the direction and magnitude of AI's distributional effects and synthesize the available evidence to date. Figure 1 summarizes the framework.[4]

We then bring the framework together in Section 4 by tracing two illustrative scenarios: one in which the mechanisms combine to produce concentrated gains, as vividly described by Meade (1964), and one in which they produce shared prosperity. These are not forecasts. Rather, they represent potential scenarios of how the same technology can generate very different distributional outcomes depending on how the key technological and economic factors evolve. Finally, we identify several empirical indicators that can reveal which path the economy is taking, describing the data sources that already track them, and highlighting remaining measurement gaps.

Our research contributes to the literature on the economic impact of AI. Current re-

[3]Gains captured as profits by firms reach household income through two channels: dividends, which raise shareholder income directly, and higher equity valuations. The latter benefit shareholders only when assets are sold, so their redistributive effect depends on who ultimately sells the appreciated assets (Fagereng et al., 2024); Section 3.2 discusses this in detail.

[4]An initial version of an interactive companion to the framework, designed to illustrate its key mechanisms and their distributional implications, is available at https://ai-distribution.org. It allows users to explore alternative scenarios for the framework's mechanisms. The tool is under active development.

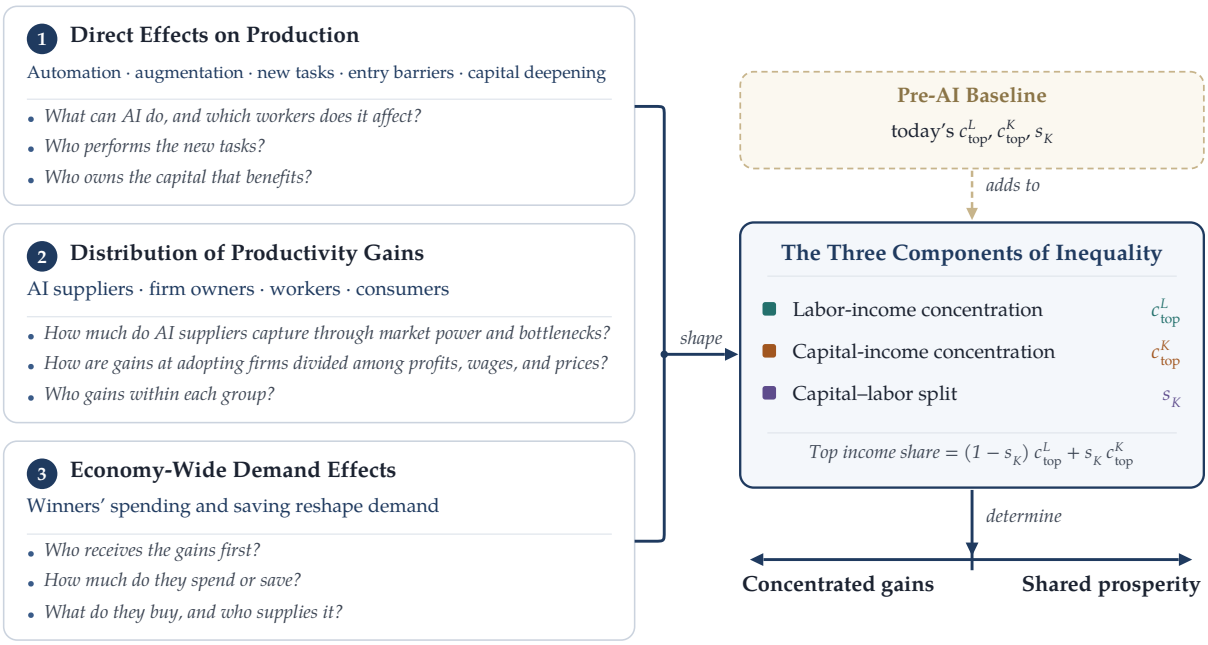


Figure 1: How AI Shapes the Distribution of Prosperity

AI's direct effects on production (1), the division of the resulting gains (2), and their spread through the economy (3) together shape the three components of inequality—labor-income concentration (c^L_{top}), capital-income concentration (c^K_{top}), and the capital–labor split (s_K). Acting on top of the pre-AI distribution, these components determine where outcomes fall between concentrated gains and shared prosperity.

search mostly focuses on three main questions. First, what effects will AI have on aggregate employment? Will AI lead to mass unemployment or not?[5] Second, what will AI do to the size of the total economic pie? How much will GDP expand, and will GDP growth explode over time?[6] Third, what are AI's existential risks? Will AI kill us all and what can we do to prevent this?[7] This paper focuses on a fourth question: how will AI's gains be distributed across people?

To answer this question, we bring together several largely separate literatures within a unified framework for analyzing AI's overall distributional effects. Our analysis of AI's direct effects on production builds on the task-based literature on technology and labor

[5]See, for example, Frey and Osborne (2017); Bessen (2019a); Acemoglu and Restrepo (2019a); Gmyrek et al. (2023); Hampole et al. (2025); Brynjolfsson et al. (2025a); Hosseini and Lichtinger (2026b).

[6]See, for example, Aghion et al. (2017); Hatzius et al. (2023); Korinek and Suh (2024); Filippucci et al. (2024); Aghion and Bunel (2024); Acemoglu (2025b); Restrepo (2025); Trammell and Korinek (2026); Jones and Tonetti (2026); Jones (2026); Korinek et al. (2026).

[7]See, for example, Bostrom (2014); Hendrycks et al. (2023); Bengio et al. (2024); Trammell and Aschenbrenner (2024); Jones (2024, 2025); RAND Corporation (2025); Bengio et al. (2026).

markets (Acemoglu and Autor, 2011; Acemoglu and Restrepo, 2018; Moll et al., 2022; Autor and Thompson, 2025; Danieli, 2026; Acemoglu et al., 2026). Our analysis of how the resulting productivity gains are distributed draws on research on market power, rent-sharing, and pass-through (Melitz and Ottaviano, 2008; Card et al., 2018; Kline et al., 2019; De Loecker et al., 2020; Autor et al., 2020a), the economics of the AI supply chain (Korinek and Stiglitz, 2019; Korinek and Vipra, 2025), the distributional effects of relative prices (Baumol, 1967; Jaravel, 2019), and the distributional effects of asset-price changes (Fagereng et al., 2024). Finally, our analysis of economy-wide demand effects draws on micro- and macroeconomic evidence on saving behavior and the composition of consumption (Buera et al., 2022; Moll et al., 2022; Hubmer, 2023; Fagereng et al., 2025; Mian et al., 2025). At each category, we connect the effects to the three inequality components. This integration allows us to examine how mechanisms typically studied in isolation combine to determine AI's overall effect on the income distribution.

We hope this framework proves useful to three main audiences. First, for researchers, it integrates mechanisms often studied in separate literatures and may help guide future research by identifying key empirical and theoretical questions about AI's distributional effects. Second, for policymakers, it provides a structured way to assess these effects, clarifying which mechanisms policy can influence and which indicators to monitor as AI diffuses. Third, for participants in the broader public debate, it provides a coherent framework for evaluating competing claims about how AI will shape the income distribution.

2 The Three Components of Income Inequality

Overview: Decomposing Income Inequality

Pre-redistribution income largely comes from two sources: labor income (wages) and capital income (returns on what people own).[8] The share of total income received by the top group (for example, the top 1%) can then be written as:

$$S_{\text{top}} \;=\; (1 - s_K)\, c^{L}_{\text{top}} \;+\; s_K\, c^{K}_{\text{top}}, \tag{1}$$

[8]Capital income consists mainly of interest, dividends, rents, business profits, and realized capital gains.

where s_K is the share of national income paid to capital (the capital share), $(1 - s_K)$ is the share paid to labor (the labor share), and c^L_{top} and c^K_{top} are the top group's shares of total labor income and of total capital income (Meade, 1964; Atkinson and Lakner, 2021; Moll et al., 2022; Gomez, 2025).[9] In words, the top income share is a weighted average of *labor-income concentration* and *capital-income concentration*, weighted by each source's *income share*. An analogous decomposition applies to other inequality measures, such as the Gini coefficient (see Appendix A.3).

Decomposing changes over time in S_{top}, following Moll et al. (2022), gives three avenues through which AI can act:

$$\Delta S_{\text{top}} = \underbrace{\Delta c^L_{\text{top}}\,(1 - s_K)}_{\text{wage concentration}} + \underbrace{\Delta c^K_{\text{top}}\, s_K}_{\text{capital concentration}} + \underbrace{\Delta s_K \left(c^{K,\text{post-AI}}_{\text{top}} - c^{L,\text{post-AI}}_{\text{top}}\right)}_{\text{capital-labor split}} \tag{2}$$

- **Labor-income concentration (Δc^L_{top}).** AI can change how unequally wages are spread across workers. This effect matters more the larger the labor share $(1 - s_K)$.
- **Capital-income concentration (Δc^K_{top}).** AI can change how concentrated capital income is across individuals. This effect matters more the larger the capital share (s_K).
- **Capital-labor split (Δs_K).** AI can change the division of income between labor and capital. This effect matters more the larger the gap between capital-income and labor-income concentration $(c^{K,\text{post-AI}}_{\text{top}} - c^{L,\text{post-AI}}_{\text{top}})$.[10]

The remainder of this section discusses each of the three components separately (Sections 2.1–2.3) and then turns to the distinction between inequality and income levels (Section 2.4).[11] For each, it explains what it is and how it evolved in the decades preceding AI. This establishes the pre-AI baseline against which the AI-driven changes analyzed in Section 3 can be evaluated.

[9] The algebraic derivation is provided in Appendix A.2.

[10] Here $c^{K,\text{post-AI}}_{\text{top}} = c^K_{\text{top}} + \Delta c^K_{\text{top}}$ and $c^{L,\text{post-AI}}_{\text{top}} = c^L_{\text{top}} + \Delta c^L_{\text{top}}$ are the post-AI concentrations.

[11] Note that AI can also affect the distribution of real purchasing power through changes in relative prices, as discussed in Section 3.2.2.

2.1 Labor Income Concentration (c^L_{top})

The term c^L_{top} is the share of total labor income accruing to a top income group. Wages are the main source of income for most households: outside the top decile, labor income far exceeds capital income. How wages are distributed therefore largely determines living standards for most people. Earnings also shape economic security, social status, and economic mobility.

Recent trends: U.S. wage inequality rose sharply after 1980. The college wage premium widened substantially, but inequality also increased among workers with similar education and experience. The rise was especially pronounced at the top: the gap between high earners and the middle widened much more than the gap between the middle and the bottom (Katz and Murphy, 1992; Autor et al., 2008, 2020b; Katz, 2025).

Drivers: Technological change was one of the main drivers of rising wage inequality. Computers complemented workers performing analytical and managerial tasks while replacing many routine clerical and production tasks, contributing to the decline of middle-paying jobs (Autor et al., 2003; Acemoglu and Autor, 2011). One estimate attributes 50–70% of changes in relative wages across demographic groups between 1980 and 2016 to automation-related task displacement (Acemoglu and Restrepo, 2022). These wage losses were particularly large in middle-income routine occupations where technology replaced workers' core skills and expertise (Autor and Thompson, 2025; Danieli, 2026). Moreover, the new, high-paying tasks created by technology became increasingly concentrated in well-paid professional and technical jobs requiring higher education (Autor et al., 2024, 2026). Weaker labor-market institutions reinforced these effects, particularly at the bottom of the distribution: the declining real value of the minimum wage reduced pay among less-educated workers, while falling unionization weakened workers' bargaining power (Cengiz et al., 2019; Farber et al., 2021; Vogel, 2025). Trade also played an important role: import competition from China (the "China shock") caused sharp and persistent declines in manufacturing employment and earnings (Autor et al., 2013, 2016, 2021).

Looking ahead: Wage inequality therefore enters the AI era high by historical standards. Past experience suggests that AI's effects will depend not only on what the technology can do, but also on how firms deploy it and on the labor-market institutions that shape workers' responses and bargaining power.

2.2 Capital Income Concentration (c^K_{top})

The term c^K_{top} is the share of capital income received at the top, including interest, rents, dividends, business income, and realized capital gains, i.e., gains on assets that are sold.[12] Capital income is considerably more concentrated than labor income, making it especially important for inequality at the top, and its importance grows as the capital share rises. The wealth underlying capital income also provides economic security, political influence, and advantages that can be passed across generations.

Recent trends: Capital income has become an increasingly important source of inequality: since 2000, most of the rise in the top 1% income share has come from capital income, particularly retained corporate profits (Piketty et al., 2018). Wealth has also become substantially more concentrated: by the early 2010s, the top 0.1% owned roughly 16–22% of U.S. wealth—up from about 7% in the late 1970s—while the share owned by the bottom 90% fell from 35% to 23% (Saez and Zucman, 2016; Smith et al., 2023).

Drivers: Capital income depends on the amount of wealth households own and the returns they earn on it, both of which are highly concentrated at the top. Wealthy households earn systematically higher returns on their assets (Fagereng et al., 2020; Smith et al., 2023), in part because their portfolios are more heavily concentrated in business equity, causing equity-price gains to accrue disproportionately to them (Kuhn et al., 2020). As a result, capital income is even more concentrated than wealth itself (Gaillard et al., 2023). Automation can further reinforce this pattern by raising demand for capital and potentially increasing returns to wealth (Moll et al., 2022). Wealthy households may also save a larger share of their incomes and therefore accumulate wealth more rapidly, further increasing wealth inequality, though there is considerable debate about this point (Straub, 2019; Mian et al., 2025; Fagereng et al., 2025).

Looking ahead: Capital income therefore enters the AI era already highly concentrated. Whether AI further increases this concentration will depend on who owns AI-related capital, how its returns are distributed, and whether opportunities to own that capital become more broadly accessible.

[12] Capital gains can be realized when an appreciated asset is sold, or unrealized while the asset is held. *Realized* capital gains are the component that matters for income and welfare; as Fagereng et al. (2024) show, their redistributive effect depends on who ultimately sells the appreciated assets. Section 3.2 discusses how AI-driven asset-price changes map into the framework.

2.3 The Division of Income Between Capital and Labor (s_K)

The term s_K is the share of national income paid to owners of capital. The remainder, the labor share $(1 - s_K)$, is paid to workers through wages and other compensation. Because capital income is more concentrated than labor income, a larger capital share generally shifts income toward those at the top of the distribution (Meade, 1964; Moll et al., 2022).

Recent trends: The labor share, once considered a relatively stable feature of economic growth, has declined substantially in recent decades. In the United States, it remained near 63% through much of the late twentieth century before falling by roughly 6 to 8 percentage points to its lowest level of the postwar period (Grossman and Oberfield, 2022; Audoly et al., 2026). Similar declines occurred in many other countries (Karabarbounis and Neiman, 2014). This decline also helps explain the growing gap between productivity and pay: across advanced economies, workers' compensation has grown slower than productivity (Suss, 2026).

Drivers: The causes of this decline remain debated (Grossman and Oberfield, 2022). One explanation is technological: as capital equipment became cheaper, firms increasingly substituted capital for labor (Karabarbounis and Neiman, 2014; Hubmer, 2023; Acemoglu and Restrepo, 2022; Moll et al., 2022). Other explanations emphasize changes in market structure: highly productive "superstar" firms with relatively low labor shares gained market share (Autor et al., 2020a), while rising markups increased the share of output captured as profits, at the expense of labor (De Loecker et al., 2020; Barkai, 2020). Part of the measured decline may also be an accounting artifact: as business activity shifted toward tax-preferred pass-through entities, some income that is effectively labor income of owner-managers was recorded as business profits (Smith et al., 2019, 2022).

Looking ahead: The labor share enters the AI era after decades of decline. AI could push it lower if it substitutes for workers or increases profits and economic rents, but it could raise the labor share if it strongly complements workers and increases their wages. Because capital income is highly concentrated, a further shift toward capital would generally increase inequality (Piketty, 2014; Autor and Kausik, 2026).

2.4 Inequality vs. Levels

Equation (1) describes how income is divided, not how much income there is. The two can move independently: if AI generates enough growth, incomes can rise throughout the distribution—including at the bottom—even as the top's share increases; conversely, an economy can be perfectly equal at a very low level of income.

To see this, consider a simple identity: total income outside the top group is $Y_{\text{bottom}} = (1 - S_{\text{top}})\, Y$, where Y is national income and S_{top} is the income share of the top group as in Equation (1). In percent changes,

$$\%\Delta Y_{\text{bottom}} \approx \%\Delta Y + \%\Delta \left(1 - S_{\text{top}}\right), \tag{3}$$

so incomes at the bottom rise whenever aggregate growth outpaces the loss of share: $\%\Delta Y_{\text{bottom}} > 0 \iff \%\Delta Y > -\%\Delta \left(1 - S_{\text{top}}\right)$.[13] For example, if the top 10% share rises from 35% to 40%, total income of the bottom 90% still rises as long as the economy grows by more than 8.3%.

Historically, the rise in inequality since 1980 described above—amid computerization, the internet, and industrial robots—was accompanied by sharp income growth at the top, while income levels in the bottom half largely stagnated (Piketty et al., 2018; Moll et al., 2022; Saez and Zucman, 2019).[14] Between 2015 and 2023, however, wage levels at the bottom rose significantly, initially driven largely by minimum-wage increases and later by post-pandemic labor-market tightness (Aeppli and Wilmers, 2022; Autor et al., 2023).

The same distinction between inequality and levels appears in the relationship between the capital–labor split and average wages. If automation generates enough growth, it can reduce the labor share while raising average wages (Acemoglu and Restrepo, 2018), and some argue that this is the likely outcome of further automation (Caselli and Manning, 2019; Autor and Kausik, 2026). But this depends on how much automation lowers costs: if cost savings per automated task are modest, output gains may be too small to offset labor displacement, causing average wages to stagnate or fall (Acemoglu and

[13] Equation (3) holds exactly in log changes, $\Delta \ln Y_{\text{bottom}} = \Delta \ln Y + \Delta \ln(1 - S_{\text{top}})$, which approximate percent changes when changes are not too large.

[14] The magnitude of these patterns is debated: using different assumptions, Auten and Splinter (2024) estimate a smaller rise in the top 1% share and find that real pretax incomes of the bottom half grew over 1979–2019 rather than stagnating. Under either set of estimates, however, top income shares rose while the bottom half's share declined.

Restrepo, 2019b; Rachel, 2026). Moreover, when capital supply is upward sloping, part of the productivity gains from automation accrues as a higher return to capital rather than higher wages; if capital supply is sufficiently inelastic, automation can therefore reduce wages (Moll et al., 2022). A historical example is "Engels' pause": during the early decades of the British Industrial Revolution, output per worker rose substantially while real wages stagnated for roughly half a century, with the gains instead accruing as higher profits (Allen, 2009).

The rest of the paper focuses on the *distribution* of income, encompassing both inequality and levels: the mechanisms in Section 3 operate first on levels—wages, prices, and profits—and the decomposition determines how those changes translate into income shares.

3 How AI Shapes the Income Distribution

The three components of income inequality in Section 2—the concentration of labor income (c^L_{top}), the concentration of capital income (c^K_{top}), and the share of income paid to capital (s_K)—provide the framework for the rest of the paper. This section examines the mechanisms through which AI may change these components and the factors that determine the direction and magnitude of those changes.

We organize the mechanisms into three interconnected categories. The first, *direct effects on production* (Section 3.1), examines how AI changes what work is performed, what work is demanded and who can supply it, and how capital-intensive production is. It includes automation, augmentation, new-task creation, changes in occupational entry barriers, and capital deepening, closely paralleling the five categories of technological change identified by Acemoglu et al. (2026).

The second category, *the distribution of the productivity gains* (Section 3.2), examines how the gains generated by AI are divided through two splits: first, between AI suppliers and adopting firms through the prices charged for AI inputs; and second, at adopting firms, among owners, workers, and consumers—as well as within each group—through higher profits, higher wages, and lower prices.

The third category, *economy-wide demand effects* (Section 3.3), examines how the spend-

ing and saving of those who initially benefit from AI reshape demand for goods and services and, in turn, for the workers who produce them.

For each mechanism, we explain what it is, identify the key questions that determine its effect on inequality, and synthesize the available evidence to date. Importantly, no single mechanism determines AI's overall effect on the income distribution; the outcome depends on how the mechanisms interact. As Section 4 illustrates, different combinations can produce very different outcomes, ranging from concentrated gains among a narrow set of firms, owners, and top earners to broadly shared prosperity through higher wages and lower prices. Keeping these two broad outcomes in mind provides a useful lens for the discussion that follows: depending on the underlying technological and economic conditions, each mechanism can push the economy toward either one—towards utopia or dystopia. Because the mechanisms are closely interconnected, the discussion includes occasional overlap and cross-references across the following subsections.

3.1 Direct Effects on Production

Table 1 previews the five mechanisms through which AI directly reshapes production. For each, it provides a brief description and the key distributional questions that organize the discussion that follows. We then examine each mechanism in detail.

3.1.1 Automation (Task-Based Labor Substitution)

Automation is technological change that allows capital—machines, software, and now AI—to perform tasks previously carried out by workers. We consider a task automated when AI performs it end-to-end with little human intervention, removing it from the worker's job bundle, along with the compensation attached to it (Acemoglu and Restrepo, 2018; Garicano et al., 2026). By contrast, we consider AI to augment a worker when it performs part of the job but the value of its output still depends substantially on the parts the worker continues to perform—for example, judgment and verification (Gans

Table 1: Summary of the Direct Effects on Production

Mechanism	Description	Key distributional questions
Automation (§3.1.1)	AI performs existing tasks end-to-end with little human intervention, substituting capital for labor.	(i) How far does automation extend? (s_K) (ii) Which workers are directly affected? (c^L_{top}) (iii) Where do displaced workers move? (c^L_{top}) (iv) Who owns the capital that benefits? (c^K_{top}) (v) Who captures the productivity gains? (c^L_{top}, s_K, c^K_{top})
Augmentation (§3.1.2)	AI raises the productivity of workers in tasks they continue to perform.	(i) Which workers experience the largest productivity gains, within and across occupations? (c^L_{top}) (ii) Who captures the productivity gains: does higher productivity translate into higher wages? (c^L_{top}, s_K, c^K_{top})
New-task creation (§3.1.3)	AI creates entirely new tasks and occupations.	(i) How much new work does AI create? (s_K) (ii) Who performs the new work? (c^L_{top}, s_K, c^K_{top})
Changes in entry barriers (§3.1.4)	AI changes the expertise required to perform tasks or enter occupations.	(i) Where does AI lower or raise entry barriers? (c^L_{top}) (ii) Who can access and use AI effectively? (c^L_{top}) (iii) Do institutions allow boundaries to change? (c^L_{top})
Capital deepening (§3.1.5)	AI increases capital used per worker through AI adoption and the process of producing AI.	(i) How much does AI increase capital intensity? (s_K) (ii) Does AI substitute for or complement labor? (s_K) (iii) Who owns the capital and captures its returns? (c^K_{top})

Notes: Each mechanism is examined in the subsection referenced beside its name, together with the main factors that determine its effect; the questions correspond one-to-one to the correspondingly numbered parts of that subsection. The symbols in parentheses indicate the inequality component(s) each question bears on most directly: the concentration of labor income (c^L_{top}), the concentration of capital income (c^K_{top}), and the share of income paid to capital (s_K), as defined in Section 2.

and Goldfarb, 2026; Garicano et al., 2026).[15] We discuss AI augmentation in Section 3.1.2. Automation is therefore the most direct way AI can reduce demand for labor in affected tasks, shift income from labor toward capital, and change which workers and skills are valued (Acemoglu and Restrepo, 2018; Korinek and Stiglitz, 2019; Moll et al., 2022).

AI automation affects inequality through three linked effects (Acemoglu and Restrepo, 2018, 2022). The first is a direct *displacement effect*: capital takes over tasks previously per-

[15] Consider a lawyer. If AI drafts routine contracts end-to-end and clients use them without the lawyer's involvement, the drafting task is automated—it leaves the lawyer's job, along with the pay attached to it. If instead AI produces the draft but its value still depends on the lawyer's review, judgment, and responsibility toward the client, AI augments the lawyer, who continues to be paid for the service.

formed by workers, shifting those tasks from labor to capital. The second is a *ripple effect*: displaced workers move into other occupations, increasing labor supply there and potentially lowering wages even for workers whose own tasks were not automated. The third is a *productivity effect*: if AI performs automated tasks more efficiently than workers, it can increase firms' productivity, allowing them to produce more output with the same inputs or the same output at lower cost. If these productivity gains are large enough, automation can raise average wages even as it displaces workers from tasks (Acemoglu and Restrepo, 2018; Caselli and Manning, 2019; Autor and Kausik, 2026). The distributional consequences of these three effects depend primarily on five questions:

(i) How far does automation extend?

(ii) Which workers are directly affected?

(iii) Where do displaced workers move?

(iv) Who owns the capital that benefits?

(v) Who captures the productivity gains?

(i) How Far Automation Goes

The extent of automation is a central determinant of its direct effect on the capital–labor split (s_K): the more tasks firms shift from workers to AI, the more income is likely to move from labor toward capital.

One task-level assessment classifies roughly 20% of U.S. tasks as highly automatable, with many more at least partly exposed.[16] Technical exposure, however, does not necessarily translate into automation: very few tasks can currently be performed end-to-end without human oversight (Eloundou et al., 2024), and whether firms automate an exposed task depends on AI's reliability, the cost of compute relative to wages, and the costs of reorganizing production and unbundling the task from the rest of the worker's job (Lashkari et al., 2026; Garicano et al., 2026). Looking further ahead, AI may also accelerate advances in AI systems and robotics, expanding the range of tasks that can be automated over time.

[16] These figures are based on LLM-generated task-level ratings of Hosseini and Lichtinger (2026a), who update the widely-used O*NET tasks scored by Eloundou et al. (2024) for automation potential on a five-level scale (T0–T4); see Appendix A.4 for details.

(ii) Which Workers Are Affected

The effect of AI automation on wage inequality (c^L_{top}) depends largely on where the most affected occupations lie in the wage distribution. Because it is difficult to predict which jobs will actually be automated, we rely on *exposure* measures, with the same caveat that these capture technical potential rather than actual displacement. Current estimates suggest that automation exposure is concentrated in information-processing and clerical tasks, such as insurance claims, data entry, proofreading, and travel services, while occupations centered on physical dexterity, in-person care, or hands-on judgment appear less exposed to current AI systems (Hosseini and Lichtinger, 2026a; Eloundou et al., 2024).

Figure 2 provides an illustrative estimate of occupations' automation exposure across the wage distribution under three scenarios for how far AI capabilities advance. A limited wave, covering only tasks AI can already perform end-to-end, would affect a relatively small share of tasks. A modest wave, covering tasks for which current AI can perform most components, would primarily affect the clerical middle, resembling earlier patterns of job polarization.[17] A broader wave of more capable AI would increasingly reach high-paying professional and managerial work. Physical and in-person occupations remain the least directly exposed under all three scenarios, although they could still be affected through ripple effects.

However, the share of automatable tasks is only part of the story. The distributional effects also depend on which tasks within an occupation AI automates. Automating an occupation's core expert tasks may erode its wage premium, whereas automating routine supporting tasks may allow workers to focus on higher-value activities (Autor and Thompson, 2025; see Section 3.1.4). Within occupations, workers whose skills better complement the remaining non-automated tasks are likely to benefit more from AI (Freund and Mann, 2026).

(iii) Where Displaced Workers Move

The distributional consequences of automation depend not only on which workers are affected, but also on where they move afterward. Highly paid professionals such as soft-

[17] The automatable share rises sharply around the 35th wage percentile (roughly $43,000 annually) because customer service representatives and general office clerks are among the largest and most automatable clerical occupations, together employing roughly five million workers.

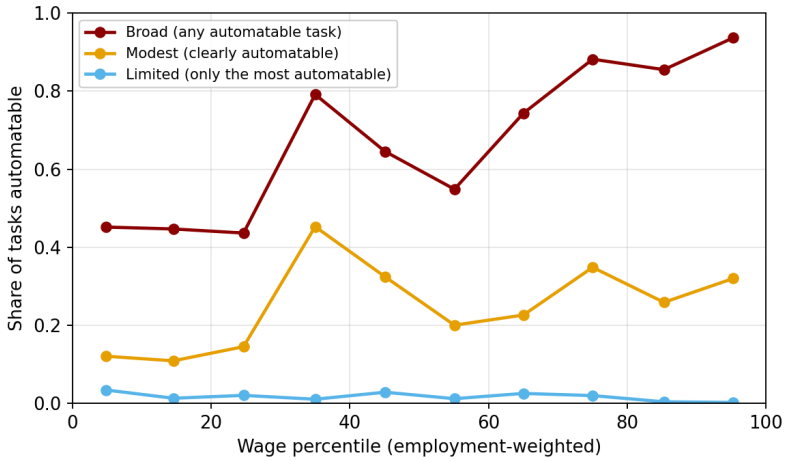


Figure 2: Estimated Share of Automatable Tasks by Occupational Wage Decile

Notes: The figure shows the employment-weighted average share of tasks within occupations estimated to be automatable by AI. These estimates are based on LLM-generated task classifications and should be interpreted as illustrative. The limited scenario includes only tasks classified as completely automatable; the modest scenario includes tasks classified as mostly automatable; and the broad scenario includes all tasks classified as at least partly automatable. Data are from Hosseini and Lichtinger (2026a), building on Eloundou et al. (2024). See Appendix A.4 for details on the data and methodology.

ware developers, financial analysts, and managers often possess skills that transfer readily across occupations, making them better positioned to adapt to displacement (Manning and Aguirre, 2026). By contrast, workers in exposed lower- and middle-paying occupations—including customer service representatives, general office clerks, secretaries, and receptionists—typically have fewer outside options, making wage losses or unemployment more likely following displacement.

Ripple effects extend the consequences to workers whose own tasks were never automated, depending on where displaced workers seek new employment. For example, if workers displaced from lower- and middle-paying clerical occupations move into lower-paying service jobs, the resulting increase in labor supply may depress wages in those occupations. These ripple effects grow with the breadth of automation: the more occupations shed workers at once, the fewer alternative destinations remain and the larger the inflows into them. Automation can therefore reduce wages even in occupations that are not directly exposed, as occurred during earlier waves of technological change (Acemoglu and Restrepo, 2020, 2022).

(iv) Who Owns the Capital That Benefits

If automation shifts income from labor to capital, its effect on capital-income concentration (c^K_{top}) depends on who owns the capital that benefits (Moll et al., 2022). AI-related capital is likely to be owned either by firms adopting AI or by firms supplying models, chips, compute, and other infrastructure. To the extent that ownership of these firms remains concentrated among wealthy households, automation is likely to increase capital-income concentration. In addition, if capital supply is not perfectly elastic, automation increases returns to wealth more broadly—especially on risky assets—disproportionately benefiting wealthy households that hold these assets (Moll et al., 2022).

(v) Who Captures the Productivity Gains

Finally, automation can generate productivity gains by allowing firms to produce more efficiently and at lower cost. As discussed in detail in Section 3.2, the distributional implications of these gains depend on who captures those gains: AI suppliers through the prices they charge, owners of adopting firms through higher profits, workers through higher wages, or consumers through lower prices. Gains concentrated in corporate profits are more likely to increase inequality because ownership is highly concentrated, whereas wage gains and lower prices may benefit a broader population—although this depends on which workers receive higher wages and which consumers buy the affected goods. This division is determined largely by market conditions, including market power, bottlenecks, worker bargaining power, and the price elasticity of demand (Section 3.2).

3.1.2 Augmentation (Labor Complementarity)

Augmentation is technological change that makes workers more productive at tasks they continue to perform. AI augments a worker when it performs part of the job but the value of its output still depends substantially on the parts the worker continues to perform—such as judgment and verification—so the worker continues to be compensated for the full job (Gans and Goldfarb, 2026; Garicano et al., 2026). At current capabilities, AI may be better suited to complement workers than to replace them outright. Unlike automation, augmentation therefore does not directly displace labor from the job. The distributional effects of AI augmentation depend primarily on two questions:

(i) Which workers' productivity rises most, within and across occupations?

(ii) Who captures the productivity gains: does higher productivity lead to higher wages?

(i) Which Workers Experience the Largest Productivity Gains

Within occupations, a growing body of experimental evidence suggests that AI often raises the productivity of less-experienced and lower-performing workers the most. Studies of customer support, professional writing, product development, and business problem-solving consistently find larger gains for these workers than for their more experienced peers, substantially narrowing performance gaps (Brynjolfsson et al., 2025b; Noy and Zhang, 2023; Dell'Acqua et al., 2025; Cruces et al., 2026). These findings suggest that AI could act as an equalizer by extending expert-level performance to less specialized workers (Autor, 2024). However, most of the evidence comes from controlled experiments on narrow, self-contained tasks. In broader workplace settings, domain expertise and judgment may remain important determinants of who benefits most from AI (Hitzig et al., 2026) and who chooses to use it (Bick et al., 2026; Handa et al., 2025; Imas and Shukla, 2026). Moreover, as AI becomes agentic—performing more of the work while humans provide direction and exercise judgment—its gains may increasingly favor experienced workers with more managerial skills (Alekseeva et al., 2026; Weidmann et al., 2025). Augmentation may also prove temporary: workers who benefit most from AI today may later become exposed to automation as AI capabilities improve (Manning, 2024).

Between occupations, task-based estimates suggest that the potential for AI augmentation rises with wages: it is largest in high-paying cognitive occupations and smallest in lower-paying physical and in-person work (Eloundou et al., 2024; Hosseini and Lichtinger, 2026a). These patterns could be reinforced if adoption is unequal. Current surveys show that AI use at work is roughly twice as common among college-educated workers as among workers without a college degree (Bick et al., 2026), and provider usage data show that actual workplace AI use is concentrated in higher-paying cognitive occupations (Anthropic, 2025; Iscenko et al., 2026). As a result, even if AI compresses productivity differences within occupations, it may widen them across occupations by raising productivity most where earnings are already highest.

(ii) Who Captures the Productivity Gains?

Higher productivity does not necessarily translate into higher wages or employment for augmented workers. Whether workers benefit depends partly on the *price elasticity of demand* for their output—how strongly demand responds when prices fall (Autor, 2015; Bessen, 2019b). If demand expands substantially, firms may increase production enough to raise demand for these workers' labor. If demand responds little, firms may instead meet existing demand with fewer workers, reducing labor demand. Tax preparation illustrates the latter case: because the number of tax returns is relatively fixed, AI assistance may make filing cheaper and faster without increasing employment or wages among tax preparers.[18]

More broadly, productivity gains from AI augmentation may accrue to AI suppliers through higher service prices, adopting firms through higher profits, workers through higher wages, or consumers through lower prices (see Section 3.2). How these gains are divided depends on market forces such as competition, workers' bargaining power, supply-chain bottlenecks, and the responsiveness of demand to lower prices.

3.1.3 New-Task Creation

New-task creation is technological change that creates new tasks and occupations. This differs from simply increasing demand for existing work: technology may expand employment in existing occupations without creating genuinely new tasks, a mechanism discussed in Section 3.3. New-task creation matters because it is the main force counteracting automation in labor demand. Whereas automation shifts existing tasks from workers to capital, new tasks can create new roles for workers and restore labor's share of income (Acemoglu and Restrepo, 2018). Historically, new work has commanded substantial wage premiums, making access to it an important determinant of wage inequality (Autor et al., 2026).

Because new tasks typically emerge only as technologies diffuse and firms reorganize production, it remains difficult to predict what new work AI will create or how impor-

[18] A classic historical example is agriculture: U.S. agricultural productivity rose dramatically during the twentieth century, but because demand for food is inelastic—people do not eat much more as food becomes cheaper—output did not expand enough to offset labor-saving productivity growth. Farm employment therefore fell even as agricultural output increased (Autor, 2015; Bessen, 2019b).

tant it will become. Nevertheless, its distributional consequences depend mainly on two questions:

(i) How much new work does AI create?

(ii) Who performs the new work?

The first determines whether new-task creation is large and fast enough to offset automation; the second, whether the gains flow to workers or to capital owners, and which workers benefit.

(i) How Much New Work AI Creates

Historically, technological change eventually generated substantial new work: more than 60% of U.S. employment in 2018 was in occupations whose job titles did not exist in 1940 (Autor et al., 2024). Some AI-related work is already emerging, including building and evaluating models, managing AI adoption within firms, creating and reviewing training data, and teaching others to use AI tools (Jaumotte et al., 2026; Adrjan, 2026). Whether this work becomes a major source of employment or stays relatively specialized alongside widespread automation remains uncertain.

(ii) Who Gets the New Work

New tasks do not necessarily create jobs for people: AI systems may perform them from the outset, or soon after they emerge, allowing production to expand with little additional demand for labor and shifting more of the gains toward capital owners (Acemoglu and Restrepo, 2018; Restrepo, 2025). When workers do perform the new tasks, the effect on wage inequality depends on who gets those jobs and how well they pay. Historically, new work has disproportionately employed younger and more-educated workers and has commanded significant wage premiums, especially when associated with new technologies (Autor et al., 2026). Since 1980, it has also become increasingly concentrated in professional and technical occupations (Autor et al., 2026; Acemoglu et al., 2026). AI could therefore widen wage inequality if new tasks remain concentrated among highly educated specialists, or narrow it if they become accessible to a broader range of workers.

3.1.4 Changes in Entry Barriers and Expertise Requirements

One important determinant of an occupation's wage is the scarcity of the expertise required to perform it (Autor and Thompson, 2025). Specialist doctors earn high wages not only because demand for their services is strong, but also because the expertise required to provide them is scarce and costly to acquire. Many other well-paid occupations similarly command premiums for scarce expertise. AI could reduce these premiums by making expert capabilities more widely available, allowing workers with less training to perform work previously reserved for specialists (Autor, 2024). However, AI could also raise entry barriers if using it effectively requires new technical, managerial, or coordination skills in occupations that did not previously require them. The distributional effects depend primarily on three questions:

(i) Where does AI lower or raise entry barriers?

(ii) Who can access and use AI effectively?

(iii) Do institutions allow occupational boundaries to change?

Together, these determine who gains access to well-paid work and how large expertise premiums remain.

(i) Where AI Changes Expertise Requirements

AI could change expertise barriers in two ways. First, it may allow workers in one occupation to perform tasks previously reserved for another: with AI assistance, paralegals might take on some tasks currently done by lawyers, while nurses perform some tasks traditionally handled by doctors (Autor, 2024). Second, it may change the expertise required to enter an occupation, allowing more or fewer workers to qualify for it.

The effect on wage inequality (c^L_{top}) depends on where these changes occur. AI could lower expertise requirements in high-paying occupations and thereby narrow wage gaps by enabling more workers to perform high-value work (Hosseini and Lichtinger, 2026a; Althoff and Reichardt, 2026). By contrast, if AI raises the technical or managerial skills required to obtain entry-level or middle-paying jobs, it could place good jobs further out of reach and widen inequality. Early evidence suggests that AI may already be raising skill requirements in some entry-level jobs (Hill, 2026; Westby et al., 2026).

(ii) Access to and Adoption of AI

Lower expertise requirements will have little effect if limited access, insufficient training, or adoption costs prevent less-skilled workers from using AI effectively. Early evidence already suggests unequal adoption: workplace use of generative AI is roughly twice as high among workers with bachelor's or graduate degrees as among those with less education (Bick et al., 2026). Moreover, schools serving more economically disadvantaged students also have lower levels of AI integration, including less teacher training, guidance, and access to AI-enabled tools (Campos and Singleton, 2026). Such gaps could limit who benefits from lower expertise barriers.

(iii) Whether Institutions Allow Occupational Boundaries to Change

Technical feasibility alone does not determine who performs a job. Licensing, credential requirements, regulation, and professional gatekeeping may preserve occupational boundaries even when AI reduces the expertise needed to perform the work (Blair and Chung, 2019). Scope-of-practice rules, for example, may prevent nurses from performing tasks traditionally reserved for physicians, while legal regulations may prevent workers without formal legal credentials from offering certain services. Employers and consumers may also continue to prefer traditionally credentialed professionals, even when less-credentialed workers can perform the work effectively with AI. Whether these institutions and preferences adapt will therefore determine whether lower expertise requirements translate into broader access to well-paid work (Autor, 2024; Acemoglu et al., 2026).

3.1.5 Capital Deepening

Capital deepening means that production uses more capital, or capital services, per worker. In the AI context, this capital includes compute, chips, data centers, software, frontier models, and cloud computing. Whereas automation and augmentation describe how AI affects workers' tasks, capital deepening describes how it affects the amount of capital used relative to labor.

AI can deepen capital in two ways. First, firms adopting AI may use more compute, software, and other AI-related capital for each worker they employ. For example, an accounting firm that keeps the same number of accountants but equips them with AI

software and cloud compute now uses more capital per worker. Second, producing AI is itself highly capital-intensive, requiring large investments in chips, data centers, and infrastructure (Goldman Sachs Research, 2025; International Energy Agency, 2025; Cottier et al., 2024). As the AI-producing sector grows, the economy may become more capital-intensive.

How capital deepening affects inequality depends primarily on three questions:

(i) How much does AI increase capital intensity?

(ii) Does AI substitute for or complement labor?

(iii) Who owns the capital and captures its returns?

(i) How Much AI Increases Capital Intensity

Through AI adoption, the scale of capital deepening depends on how widely and intensively firms use AI-related capital and on its cost relative to labor. Through AI production, it depends on how capital-intensive the AI sector remains and how large that sector becomes relative to the rest of the economy. Lower AI costs, broader adoption, and faster growth of AI production would therefore increase the use of capital per worker, making the remaining distributional questions more consequential.

(ii) Whether AI Capital Substitutes for or Complements Labor

More capital per worker does not necessarily mean that a larger share of income goes to capital. If it complements labor strongly enough, it can raise workers' productivity and wages, and labor's share may increase rather than fall. However, if AI capital substitutes for labor—for example, by automating tasks previously performed by workers—it can reduce labor's share of income. Studies of earlier waves of capital deepening reach different conclusions about whether capital primarily substituted for or complemented labor (Karabarbounis and Neiman, 2014; Hubmer, 2023; Lawrence, 2015).

Whether AI capital will be more substitutable for or complementary to labor than earlier technologies remains uncertain. Much AI development targets tasks currently performed by people, potentially making AI capital especially substitutable for labor (Brynjolfsson, 2022; Acemoglu et al., 2026). Yet current systems still need substantial human

verification and judgment, making them better suited in many settings to complement workers than to replace them outright (Autor, 2024; Rabanser et al., 2026; Burn-Murdoch and O'Connor, 2026; Iscenko et al., 2026). Which force dominates will depend on AI's capabilities and how firms deploy it.

(iii) Who Owns the Capital and Captures Its Returns

If capital deepening raises the capital share, its distributional effect depends on who receives the additional capital income. The returns to AI-related capital are likely to accrue to adopting firms and to suppliers of chips, compute, models, and infrastructure. Because ownership of businesses and financial assets is highly concentrated, these returns are likely to accrue disproportionately to wealthy households, increasing capital-income concentration (c^K_{top}) (Piketty, 2014; Saez and Zucman, 2016; Moll et al., 2022). As discussed in Section 3.2.1, this effect is likely to be larger when bottlenecks and market power in the AI supply chain allow a few firms to capture a large share of the gains. By contrast, if competition, open models, and falling costs lower barriers for smaller businesses, AI's gains could diffuse broadly across downstream adopters and consumers rather than driving further concentration (Nordhaus, 2004; Firooz et al., 2026).

3.2 Who Captures the Productivity Gains?

As discussed in the previous section, AI can generate substantial productivity gains for adopting firms, for example by automating or augmenting workers' tasks.[19] This section examines how those gains are divided and the market forces that shape their allocation. We trace the gains downstream through two splits. First, Section 3.2.1 examines how AI suppliers—such as providers of models, compute, chips, and other key inputs—may capture part of the gross gains through the prices they charge, reducing the amount that remains for adopting firms. Second, Section 3.2.2 examines how the gains remaining after payments to AI suppliers are divided among adopting-firm owners, workers, and consumers—and within each group—through higher profits, higher wages, and lower prices. The allocation in both splits depends less on AI's technical capabilities than on

[19] By a productivity gain, we mean that AI allows a firm to produce more output with the same inputs, the same output with fewer inputs, or some combination of the two.

market forces such as competition, workers' bargaining power, supply-chain bottlenecks, and the responsiveness of demand to lower prices. Figure 3 summarizes these two splits and the market forces governing them.

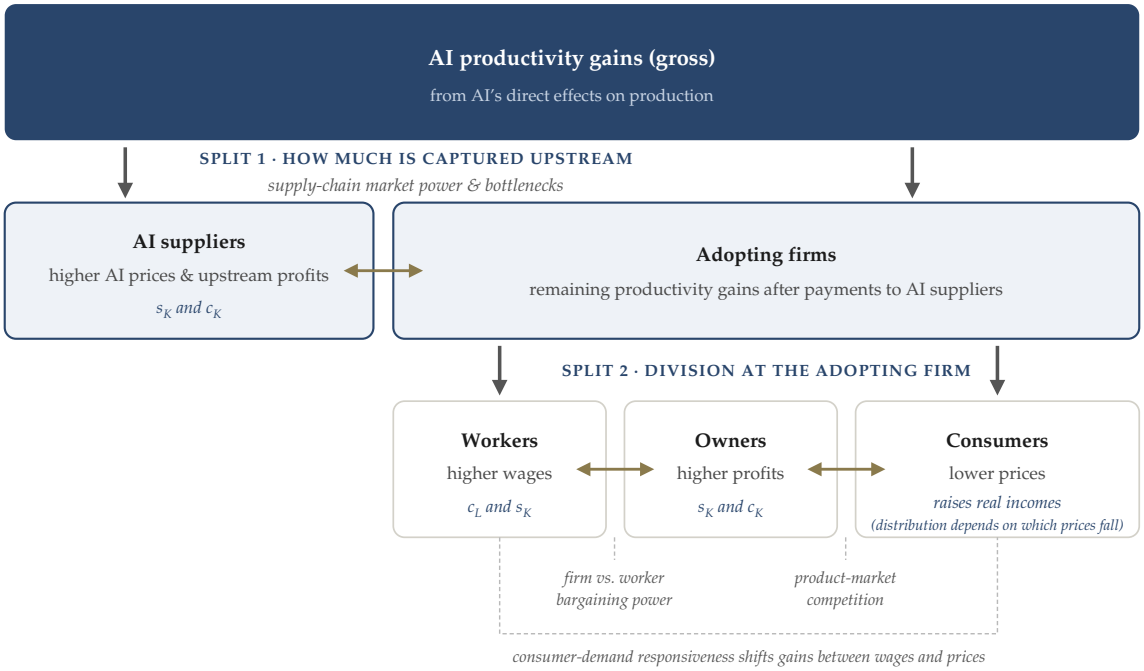


Figure 3: How AI's Productivity Gains are Divided

Notes: AI suppliers first capture part of the gross gains through the prices they charge (Split 1, shaded boxes). The remaining gains are then divided at the adopting firm among workers, owners, and consumers (Split 2, unshaded boxes). Double-headed arrows mark the boundaries that each market force shifts. The symbols in each box indicate the inequality components most directly affected, as defined in Section 2. Box widths are illustrative and do not represent estimated shares.

3.2.1 Upstream Capture in the AI Supply Chain

When a firm adopts AI, it pays for models and other AI services supplied by companies such as OpenAI or Anthropic. The price it pays reflects both costs incurred throughout the AI supply chain—including chips, memory, and data centers—and any markups or scarcity rents captured by suppliers (Korinek and Stiglitz, 2019; Korinek and Vipra, 2025). That price is a key determinant of where AI's productivity gains accrue: higher prices allow suppliers to capture a larger share of the gains, leaving less for adopting firms. We

focus on two factors that shape how much suppliers can charge: (i) market power in the AI supply chain and (ii) bottlenecks in the supply of key inputs.

(i) Market Power in the AI Supply Chain

The AI supply chain is highly concentrated, with individual firms dominant at several key stages. The most advanced AI chips are designed largely by Nvidia, fabricated by TSMC using lithography equipment supplied primarily by ASML, and paired with specialized memory supplied by three firms (Sastry et al., 2024). Such concentration gives suppliers substantial pricing power. Nvidia's gross margin, for example, has recently been around 75% (NVIDIA Corporation, 2026). When suppliers can sustain margins of this magnitude, they capture a larger share of AI's productivity gains as profits, leaving less for adopting firms, workers, and consumers.

(ii) Bottlenecks in the AI Supply Chain

Another factor determining how much AI suppliers can charge is the extent of bottlenecks along the supply chain. A bottleneck arises when demand for an input grows faster than the industry's ability to produce it, putting upward pressure on its price. This dynamic is already visible throughout the AI supply chain. The main current constraints include advanced packaging, such as TSMC's CoWoS, and high-bandwidth memory, with electricity generation and grid capacity becoming increasingly important bottlenecks as well (Somala, 2026; International Energy Agency, 2025). These inputs scale slowly: new fabrication and packaging lines, power plants, and grid connections take years to build, while demand for AI compute has been growing rapidly (Somala, 2026). These constraints can allow upstream suppliers to capture some of the gains from cheaper, more capable AI through higher input prices and scarcity rents, limiting pass-through to adopters.

Distributional Implications of Upstream Capture

Whether driven by market power or bottlenecks, upstream capture has important implications for inequality. Most of the gains captured upstream are likely to accrue as profits, increasing the capital share (s_K). Because corporate equity is disproportionately owned by wealthy households, these profits are also likely to increase capital-income

concentration (c^K_{top}). Some gains are also likely to accrue to workers employed by AI suppliers, but because these workers are among the highest-paid in the economy (Semiconductor Industry Association, 2021; NVIDIA Corporation, 2025; Bhaimiya, 2025), wage gains concentrated among them could increase labor-income concentration (c^L_{top}). How important these channels remain will depend on how competition evolves (Korinek and Vipra, 2025), including whether cloud providers' in-house chips and new entrants erode incumbent suppliers' market power, and on how quickly current bottlenecks ease.

3.2.2 How Adopting Firms Divide the Gains

After paying AI suppliers for their services, adopting firms are still likely to retain substantial productivity gains.[20] These gains can then be shared with workers through higher wages, passed on to consumers through lower prices, or retained by owners as higher profits. How these gains are divided turns on three margins, each shaped primarily by a different market force: (i) the wage-profit split, shaped by labor-market power; (ii) the price-profit split, shaped by product-market competition; and (iii) the wage-price split, shaped by the responsiveness of consumer demand to lower prices. Put differently, the allocation of productivity gains is a question of pass-through: how much firms pass on to workers through higher wages and to consumers through lower prices, rather than retain as profits.[21]

(i) Labor-Market Power: the Profit–Wage Split

Whether an adopting firm's AI gains reach its workers or stay with owners depends on the balance of labor-market power between them. When workers are hard to replace, can easily move to rivals, and have good outside options, the firm must share the gains to keep them. When labor markets are concentrated or search frictions are substantial,

[20] Note that AI could also affect firms that never adopt: adopters may gain market share at rivals' expense, while downstream firms could benefit from cheaper inputs. These spillovers operate through the same margins—wages, profits, and prices—discussed in this section.

[21] Our discussion follows models of competition with variable markups in the tradition of Melitz and Ottaviano (2008). In these models, the prices firms can charge are constrained by the alternatives available to their customers. The fewer and weaker a firm's competitors, the smaller the share of a cost reduction it must pass on through lower prices (for empirical evidence, see Amiti et al., 2019). The same logic applies in the labor market: the wage a firm must pay is disciplined by workers' outside options, so the fewer employers competing for its workers, the less of a productivity gain it must hand over in higher wages (Berger et al., 2022).

firms can retain more of the gains as profit (Caldwell et al., 2026; Kline, 2025). Unions can further strengthen workers' leverage by pressing firms to pass more of the gains on to workers through higher wages (Jäger et al., 2025). The long decline of unionization in the United States may therefore leave workers less well positioned to capture the gains from AI (Farber et al., 2021). Moreover, AI may itself shift the balance of power further toward employers: AI-enabled monitoring, algorithmic management, and predictive wage-setting can reduce workers' bargaining leverage and allow firms to retain a larger share of the gains (Davis, 2024).

(ii) Product-Market Power: the Price–Profit Split

Whether the gains reach consumers as lower prices depends on competition in the product market. Where competition is strong, a larger share of cost savings is passed through into lower prices; where firms have market power, they can retain more of the savings as profit. Rising product-market markups in recent decades illustrate how market power can tilt productivity gains toward profits rather than consumers (De Loecker et al., 2020). Whether AI reinforces this tilt toward profits or reverses it will depend on how competitive the sectors experiencing its biggest productivity gains prove to be—and on how AI itself reshapes competition through its effects on entry and innovation.

In addition to changing the price–profit split within firms, AI can affect the aggregate split by reallocating economic activity across firms. If AI's gains accrue disproportionately to dominant "superstar" firms, the aggregate profit share can rise as economic activity shifts toward large, high-markup firms with relatively low labor shares (Autor et al., 2020a).

(iii) Product-Demand Responsiveness: the Wage–Price Split

Beyond market power, the effect of productivity gains on wages and prices depends on how consumer demand for the firm's output responds. If demand is sufficiently elastic, productivity gains can generate a large expansion of output, so firms demand more labor, and workers can gain through higher employment and pay. However, if demand is inelastic, output expands little, so firms may need fewer workers to produce a given level of output, and labor demand falls (Autor, 2015; Bessen, 2019b; see the detailed discussion in Section 3.1.2). In this case, a larger share of the gains may be passed to consumers

through lower prices. Demand responsiveness therefore shapes the wage–price split.

Distributional Implications of the Profit-Wage-Price Split

The distributional effects of AI-driven productivity gains at adopting firms depend on two questions: how the gains are divided among firm owners, workers, and consumers; and who within each group receives them.

The distributional effects of productivity gains captured as firm profits require understanding how these gains translate into personal income through two channels: dividends and capital gains. First, higher dividends directly raise shareholder income, increasing the capital share (s_K). Moreover, if these gains accrue disproportionately to wealthy households, they increase capital-income concentration (c^K_{top}). Second, higher profits can be capitalized into equity valuations, generating unrealized capital gains. As Fagereng et al. (2024) show, these valuation gains should not simply be treated as income accruing to existing shareholders: their redistributive effect depends on the net present value of asset *sales* and therefore on who ultimately sells the appreciated assets. To the extent that wealthy households are net sellers of appreciated equities (as found by Fagereng et al., 2024), this channel also tends to redistribute toward the top, increasing capital-income concentration (c^K_{top}).

Gains that reach workers instead increase the labor share. Their effect on wage inequality (c^L_{top}), however, depends on which workers receive them. Workers with greater bargaining power and those in sectors with more elastic demand are likely to capture a larger share of the gains. High earners often have greater bargaining power because they possess harder-to-replace skills and better outside options (Kline et al., 2019). Whether high- or low-paid workers are concentrated in sectors with more elastic demand remains an open empirical question.

Gains passed through as lower prices raise real purchasing power. Because lower-income households consume a larger share of their income, broad-based price declines can benefit them proportionally more. However, the distributional implications depend on *which prices* fall and who consumes those goods. AI may reduce prices primarily in cognitive and digital sectors where it can generate large productivity gains, while labor-intensive services such as healthcare, childcare, and education—where productivity grows slowly—could become relatively more expensive, reflecting "Baumol's cost dis-

ease" (Baumol, 1967).[22] If AI-exposed goods are disproportionately consumed by higher-income households while essential services remain expensive, these relative-price effects could widen real-income inequality unless AI eventually raises productivity in those services as well.[23]

3.3 Economy-Wide Demand Effects

The previous sections examined how AI reshapes production and who captures the productivity gains. But those gains do not stop where they first accrue. Higher wages and profits raise household income, while lower prices raise households' purchasing power. Households then spend or save these gains. Spending creates demand for goods and services, and therefore for the workers who produce them, while saving accumulates as financial assets and may finance investment or borrowing elsewhere. As a result, productivity gains in one part of the economy can raise labor demand in another, so the ultimate winners and losers need not be the workers whose tasks AI directly affects. This demand channel had distributional effects in previous technological transitions: as productivity growth made households richer, they shifted spending toward services such as healthcare, restaurant meals, and financial services, and employment moved from farms and factories toward those services (Autor, 2015; Buera et al., 2022; Hubmer, 2023).

The distributional consequences of these economy-wide effects depend on three questions: (i) who receives the initial gains? (ii) what share do they spend vs. save?; and (iii) what do they spend on, and who meets that demand? Figure 4 summarizes this chain.

(i) Who Receives the Gains?

The demand channel begins with the households that ultimately receive AI's gains. Wage gains accrue to particular workers, profits disproportionately to wealthy shareholders, and lower prices to households that consume the affected goods. This determines whose income or purchasing power rises and, therefore, whose spending and saving decisions shape subsequent demand.

[22] This pattern is already visible across rich economies: prices of tradable goods have fallen sharply while those of essential services have soared, raising the share of middle-class income spent on essentials from roughly one-third to one-half (Burn-Murdoch, 2025).

[23] The price benefits of product innovation in recent decades accrued disproportionately to high-income households, generating sizable inflation inequality (Jaravel, 2019, 2021; Chakrabarti et al., 2026).

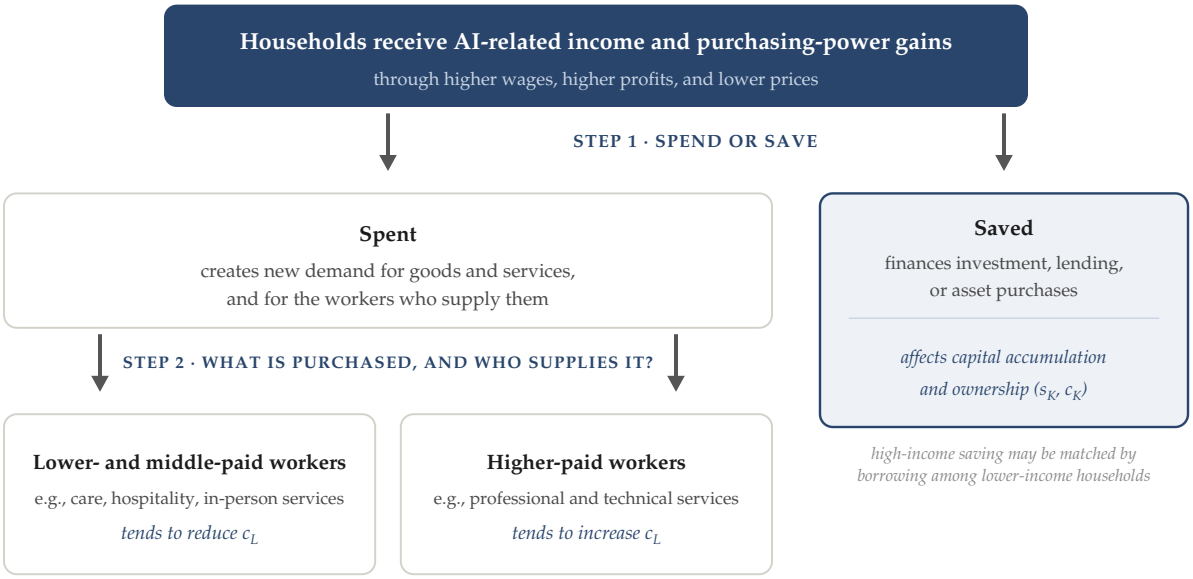


Figure 4: How AI's Gains Translate into New Demand

Notes: The figure follows AI's productivity gains after their initial division among workers, owners, and consumers (Section 3.2). Households spend or save the additional income (Step 1). Spending creates new demand for goods and services, and for the workers who supply them; the distributional effects then depend on whose labor meets the new demand (Step 2). Saving generates less immediate consumption demand and instead builds financial assets. Depending on how those savings are intermediated, they may finance investment, borrowing, or asset purchases. The figure depicts the first round of these effects; in practice, the process continues as the resulting incomes are themselves spent or saved, and as savings finance investment and spending by others.

(ii) Do They Spend or Save?

The second question is how much of these gains is spent rather than saved. If AI's gains accrue primarily to individuals who save a larger share of their income,[24] more of the gains are saved rather than spent, and their effect on demand depends on what that saving finances. In principle, it can fund productive investment, which itself generates demand for workers. Over recent decades, however, the steep rise in saving by high-income households was matched by greater borrowing among low- and middle-income households and the government (Mian et al., 2025), raising their debt rather than

[24]Evidence on how saving rates vary with income and wealth is mixed: some studies find that richer households save a larger share of their income (e.g., Dynan et al., 2004; Straub, 2019), while others do not (e.g., Fagereng et al., 2025).

financing productive investment.[25] Moreover, some of this additional saving may flow into existing assets, raising asset prices and reinforcing wealth concentration at the top (Auclert et al., 2025; Gomez and Gouin-Bonenfant, 2024; Kuhn et al., 2020).

(iii) What Do They Spend Their Income On, and Who Meets That Demand?

The third question concerns the composition of the additional spending. Richer households devote larger shares of their budgets to labor-intensive services, including restaurants, travel, and education, and smaller shares to necessities such as utilities and food consumed at home (Hubmer, 2023). At the same time, their spending tilts toward skill-intensive services such as medicine, education, finance, and business services (Buera et al., 2022).

These spending patterns determine which sectors expand and, in turn, which workers experience stronger labor demand. If AI's gains accrue primarily to high-income households, demand may tilt toward skill-intensive services and highly educated workers. If spending instead expands in labor-intensive, in-person services such as education, care, hospitality, and food service, gains are more likely to reach low- and middle-income workers.

Distributional Implications of the Economy-Wide Demand Effects

Taken together, these margins determine how AI's initial gains propagate through the broader economy. When gains accrue disproportionately to households with a low propensity to consume, a larger share is saved. If those households are also wealthy, these savings can reinforce wealth accumulation at the top and may finance greater borrowing by low- and middle-income households, potentially increasing capital-income concentration (c^K_{top}) (Mian et al., 2025). By contrast, when AI's gains accrue more broadly to low- and middle-income households, a larger share is likely to be spent, generating stronger demand for goods, services, and labor. Broader receipt of the gains may also spread saving and asset accumulation across a wider range of households, potentially reducing capital-income concentration over time.

For the portion that is spent, the distributional implications depend on its composi-

[25] Debt-financed spending can sustain demand in the short run, but it is likely to depress it further ahead (Auclert and Rognlie, 2018; Mian et al., 2021).

tion. If additional spending shifts toward relatively labor-intensive services, it is likely to support the labor share $(1 - s_K)$ (Hubmer, 2023). Which services expand also matters: if demand rises mainly for skill-intensive services—such as medicine, finance, and legal and business services—it is likely to increase demand for highly educated workers and widen labor-income inequality (c^L_{top}) (Buera et al., 2022). By contrast, spending directed toward in-person and middle-wage services may spread labor-market gains more broadly across low- and middle-income workers.

4 Synthesis: Concentration or Shared Prosperity

Section 3 examined the main mechanisms through which AI may shape the income distribution. Their aggregate distributional effect will depend on how they combine. This section brings the framework together in two steps. Section 4.1 develops two polar scenarios: one in which all the mechanisms reinforce concentrated gains (Section 4.1.1), and another in which they reduce inequality and generate shared prosperity (Section 4.1.2). These are not forecasts, and the actual outcome will almost certainly combine elements of both. Rather, they illustrate how the same technology can produce sharply different distributional outcomes and clarify the forces that will determine which path the economy follows. Figure 5 summarizes the two scenarios. Section 4.2 then identifies the indicators that can reveal which direction the economy is taking, the data available to track them, and the main remaining measurement gaps.

4.1 Two Polar Scenarios

4.1.1 A World of Concentrated Gains

In the first world, the direct effects on production (Section 3.1) largely favor the top. For high-paid professionals, AI automates the routine parts of the job and augments the core: it drafts, summarizes, and analyzes while the expert directs and judges, raising their productivity and the value of their expertise. For many middle-paying occupations, by contrast, AI automates the very tasks that justified their wage premiums. New tasks emerge only slowly, and those that do are performed mainly by highly skilled workers or by AI

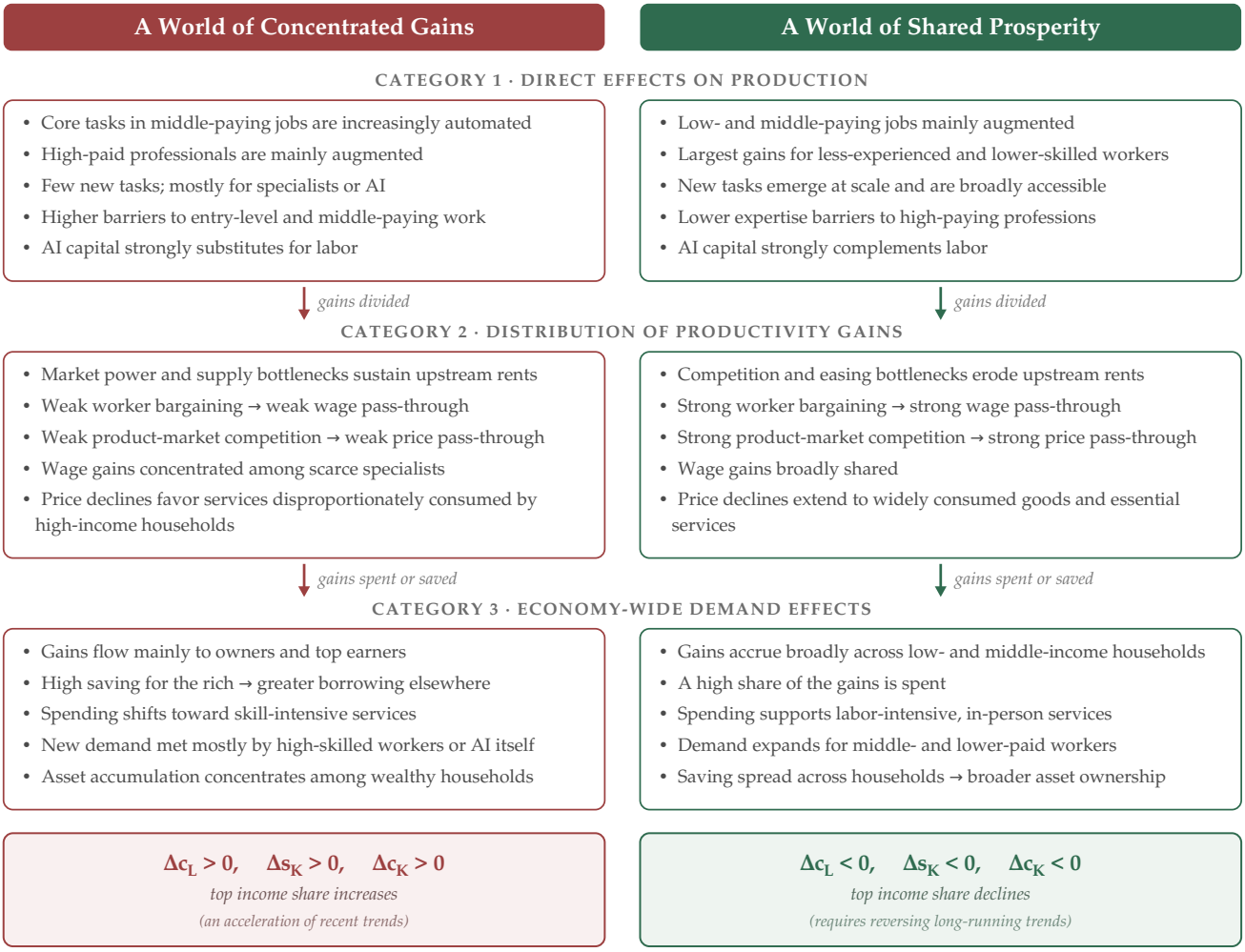


Figure 5: Summary of the Two Polar Scenarios

Notes: Each column traces one polar scenario through the three categories of the framework. These scenarios are illustrative rather than forecasts; actual outcomes may combine elements of both. See Section 4.1 for details.

itself. At the same time, using AI effectively requires new technical and managerial skills, making entry into well-paid occupations more difficult rather than easier. Production also becomes increasingly capital-intensive, with firms substituting compute for labor.

In this scenario, a large share of the resulting productivity gains (Section 3.2) is captured within the AI supply chain. Persistent bottlenecks and entry barriers in the supply of chips, compute, and energy allow AI suppliers to charge high prices and capture

substantial economic rents. Adopting firms with market power retain much of what remains as profit, while workers with weak bargaining power receive relatively little. The wage gains that do occur accrue mainly to the scarce specialists who build, manage, and direct AI systems. Meanwhile, AI lowers prices primarily in cognitive and digital services—such as legal and financial services—which are consumed disproportionately by higher-income households. In contrast, labor-intensive services such as childcare, healthcare, and education become relatively more expensive because productivity grows more slowly in those sectors (Baumol's cost disease).

The gains therefore accrue disproportionately to shareholders and high-income workers and are largely saved rather than spent (Section 3.3). This reinforces wealth accumulation at the top and may finance greater borrowing by low- and middle-income households. The spending that does occur is concentrated in skill-intensive services, further increasing demand for highly skilled workers. Some of the new demand is also met by AI itself, generating additional returns for AI suppliers and capital owners.

In this world, all three avenues from Equation 2 reinforce one another: wage inequality rises ($\Delta c^L_{\text{top}} > 0$), income shifts from labor to capital ($\Delta s_K > 0$), and capital income becomes even more concentrated ($\Delta c^K_{\text{top}} > 0$). Importantly, this outcome does not require a break with history. It extends the trends documented in Section 2: rising wage inequality, a declining labor share, higher markups, and growing wealth concentration. Concentrated gains are therefore not a departure from the recent past, but an acceleration of it.

Even in this world, however, the implications for income levels remain open. With modest aggregate gains, incomes outside the top could stagnate or even fall, resembling the post-1980 pattern; with sufficiently large gains, incomes could rise even at the bottom despite sharply rising inequality (Equation (3)).

4.1.2 A World of Shared Prosperity

In the second world, AI's direct effects on production (Section 3.1) broadly favor workers, especially those in the middle and lower parts of the wage distribution. Reliability constraints—or deliberate deployment choices—limit end-to-end automation, and AI is used primarily to augment workers rather than replace them. It takes over routine tasks

while helping workers perform the judgment-intensive parts of their jobs, with the largest productivity gains accruing to less-experienced and lower-skilled workers. AI also lowers barriers to scarce expertise, enabling less-trained workers to perform tasks previously reserved for specialists and reducing wage premiums in high-paying professions. Together, these effects narrow productivity gaps and expertise premiums both within and between occupations. Moreover, new tasks emerge at scale and are accessible to a broad range of workers rather than concentrated among a small group of highly educated specialists.

In this scenario, the resulting productivity gains (Section 3.2) are broadly shared. Competition among AI providers, falling model prices, open alternatives, and easing compute bottlenecks erode upstream rents. Competitive product markets pass much of the remaining gains on as lower prices, including for the goods and services that dominate ordinary households' budgets. At the same time, workers throughout the wage distribution have enough bargaining power to capture a meaningful share through higher wages.

These gains therefore accrue primarily to low- and middle-income households, and a large share is spent rather than saved (Section 3.3). This spending supports demand for goods and services and for the workers who provide them, particularly in labor-intensive, in-person sectors—such as care, hospitality, and parts of education—where AI remains less able to substitute for labor and many lower- and middle-paid workers are employed. The income that is saved, meanwhile, is spread across a broad range of households, allowing more families to accumulate financial assets and potentially broadening capital ownership, while those savings may also finance productive investment.

In this world, all three components of the framework move toward a more equal distribution: wage inequality narrows ($\Delta c^L_{\text{top}} < 0$), the labor share rises ($\Delta s_K < 0$), and capital income becomes less concentrated ($\Delta c^K_{\text{top}} < 0$). The top income share therefore declines. Unlike the concentrated-gains scenario, this outcome would require a reversal of several long-running trends: the automation of middle-income work, weakening competition, declining worker bargaining power, increasingly concentrated capital ownership, and unequal access to technology-driven productivity gains. Shared prosperity is possible, but it would require a substantial departure from the economic trajectory of recent decades.

4.2 What to Watch: Indicators of the Direction

Identifying which distributional path is emerging has important policy implications. This section therefore focuses on the indicators that can reveal the direction the economy is taking. Section 4.2.1 organizes these indicators around the three categories of the framework: direct effects on production, the division of productivity gains, and economy-wide demand effects (see Table 2 for a summary). Section 4.2.2 then reviews the main data sources and empirical initiatives available to track them, while Section 4.2.3 identifies the most important remaining measurement gaps.

4.2.1 Indicators Across the Framework

Indicators of the Direct Effects

The first set of indicators concerns how AI is used in production (Section 3.1). Provider usage data can reveal whether AI primarily augments workers or performs tasks end-to-end, and which occupations are most exposed. Adoption data can show whether AI use spreads across education levels, occupations, industries, and firms or remains concentrated among highly educated workers and large firms. Labor-market outcomes then capture the consequences, including changes in wages, employment, and hiring in highly exposed occupations, especially clerical and entry-level professional work, and within adopting firms. They can also reveal which workers experience the largest productivity gains.

Worker-transition data indicate whether displaced workers move into similarly paid jobs or fall down the wage distribution. Job postings can reveal whether genuinely new occupations and tasks are emerging, how well they pay, and how broadly accessible they are. They can also indicate whether employers are raising or lowering skill requirements in existing jobs. Finally, the cost of using AI to perform tasks end-to-end relative to wages provides a direct measure of firms' incentives to automate rather than augment workers.

Indicators of Who Captures the Gains

The second set of indicators tracks how productivity gains are divided (Section 3.2). Upstream, the margins and market shares of AI suppliers can reveal how much of the

gains is captured as rents, while the prices and capacity of key inputs—compute, advanced packaging, memory, and electricity—show how binding supply bottlenecks remain. Financial markets add a forward-looking but noisy signal: the valuations of AI suppliers reflect expected future profits, which combine expectations about market growth, margins, and the persistence of rents and bottlenecks.

Downstream, markups and concentration in AI-adopting industries indicate the market power that shapes how the remaining gains are divided among firms, workers, and consumers. Changes in prices and wages can then reveal whether cost reductions are passed on to consumers and whether worker compensation tracks productivity in adopting firms. The responsiveness of demand in AI-exposed sectors can further indicate whether productivity gains translate primarily into higher output and employment or into lower prices. Finally, inflation measured across different household consumption baskets can reveal whose cost of living benefits most from AI-driven price declines (Jaravel, 2019).

Indicators of Economy-Wide Demand Effects

The third set of indicators tracks how AI's gains spread through the economy (Section 3.3). Household saving and debt can show whether the gains are primarily spent, accumulated as wealth, or used to finance borrowing elsewhere. Individual-level consumer data and surveys can help estimate how households receiving AI-related gains spend or save them and which goods and services they purchase. Changes in the sectoral composition of employment can then show where the resulting demand lands and which workers ultimately benefit, including whether employment growth remains concentrated in professional services or extends to in-person and middle-wage occupations.

The Scoreboard

Ultimately, the clearest evidence comes from the three inequality components themselves. Distributional national accounts track labor-income concentration (c^L_{top}), the capital share (s_K), and capital-income concentration (c^K_{top}) (Piketty et al., 2018). These are lagging indicators: they are measured relatively infrequently and may respond only gradually to technological change. Yet they provide the ultimate test of which distributional path is emerging.

Table 2: Indicators of Concentrated Gains and Shared Prosperity

Indicator	Signal of concentrated gains	Signal of shared prosperity
Direct effects on production (Section 3.1)		
Automation versus augmentation in AI use	Increasing end-to-end automation, especially in middle-paying and entry-level work	AI remains primarily assistive and raises workers' productivity
Adoption across workers and firms	Use remains concentrated among highly educated workers and large firms	Use spreads across education levels, occupations, industries, and firm sizes
Productivity gains across workers	Gains favor experienced, high-performing, or managerial workers	Gains are largest for less-experienced and lower-performing workers
Wages, employment, and hiring in exposed work	Falling wages and employment in entry-level and lower-skilled jobs	Rising employment and wages for lower-skilled and less-experienced workers
Worker transitions after displacement	Displaced workers move into lower-paid work or leave employment	Displaced workers move into similarly paid or better-paid work
New tasks, occupations, and skill requirements	Few new roles emerge, and access requires advanced technical or managerial skills	New roles emerge at scale and are accessible to a broad range of workers
Cost of end-to-end AI automation relative to wages	Falls below labor costs across a growing range of tasks	Remains high for reliable autonomous performance
Division of productivity gains (Section 3.2)		
AI suppliers' margins and market shares	Persistent high margins and concentration among a small number of suppliers	Entry, competition, and open models reduce margins and concentration
Prices and capacity of key AI inputs	Persistent shortages and scarcity rents in compute, memory, packaging, or energy	Capacity expands and input prices decline
Market valuations of AI suppliers	Valuations imply expectations of persistent margins, bottlenecks, and rents	Valuations imply expectations of stronger competition and declining margins
Profits, wages, and prices in adopting firms	Gains appear mainly as higher profits, with little wage or price pass-through	Gains are shared through higher wages and lower consumer prices
Wages relative to productivity in adopting firms	Productivity rises substantially faster than worker compensation	Worker compensation rises alongside productivity
Output and employment responses in AI-exposed sectors	Output responds weakly and employment declines	Output expands enough to sustain or increase employment
Price changes across household consumption baskets	Price declines favor goods and services consumed disproportionately by high-income households	Price declines extend to widely consumed goods and essential services
Economy-wide demand effects (Section 3.3)		
Household spending, saving, and debt	Saving and wealth rise at the top while borrowing rises below	Spending and asset accumulation increase broadly across households
Composition of household spending	Additional spending is concentrated in skill-intensive services	Spending expands across labor-intensive, in-person, and broadly consumed services
Sectoral composition of employment growth	Employment growth is concentrated in high-skill professional occupations	Growth extends to middle-wage and in-person occupations
Inequality outcomes (Section 2)		
Labor-income concentration (c_L)	Wage inequality rises	Wage inequality stabilizes or declines
Capital share (s_K)	Income shifts from labor toward capital	The labor share stabilizes or rises
Capital-income concentration (c_K)	Capital income becomes more concentrated	Capital income becomes less concentrated

Notes: The columns describe directional signals rather than numerical thresholds. No single indicator is decisive, and indicators may temporarily move in different directions. The strongest evidence will come from their joint movement across the three categories of the framework and the three inequality outcomes.

4.2.2 Existing Data and Measurement Initiatives

Monitoring these indicators would not begin from scratch. Existing official statistics already provide much of the necessary foundation. BLS data track employment, wages, and prices; household surveys such as the CPS and CEX support long-standing research on wage inequality and consumption (Katz and Autor, 1999; Autor et al., 2020b); the Census Bureau's Business Trends and Outlook Survey provides frequent estimates of firm-level AI adoption (U.S. Census Bureau, 2026); and distributional national accounts measure the three inequality components themselves (Piketty et al., 2018).

A complementary measurement ecosystem has emerged alongside these official sources. Companies such as Lightcast and Revelio Labs compile near-real-time data from job postings and online worker profiles. AI providers publish usage data through initiatives such as the Anthropic Economic Index, OpenAI's usage studies, and Google's ATLAS (Anthropic, 2025; Chatterji et al., 2025; Iscenko et al., 2026). Epoch AI collects systematic data on compute, hardware, and the AI supply chain (Somala, 2026), while specialized surveys track AI adoption and use among workers and firms (Bick et al., 2026; Lane et al., 2023).

Efforts to combine these sources are beginning to provide more integrated views of AI's economic effects. The Stanford Digital Economy Lab's AI Economic Indicators bring together payroll, survey, and macroeconomic data in regularly updated dashboards. Its Canaries Dashboard, for example, uses ADP payroll records to track employment by age and AI exposure (Stanford Digital Economy Lab, 2026; Brynjolfsson et al., 2025a). Other work links job postings to résumé histories to study how firm-level AI adoption changes hiring across seniority levels (Hosseini and Lichtinger, 2026b). Together, these initiatives illustrate how multiple data sources can be combined to track AI's economic and distributional effects in near real time.

4.2.3 Remaining Measurement Gaps

Despite this progress, several central components of the framework remain poorly measured or are not measured in ways that can be directly linked to AI. Regarding AI's direct effects on production, the main gaps include the cost of automating specific tasks, the creation of genuinely new tasks, and the distribution of productivity gains across workers. For the division of productivity gains, the main gaps include market power and rents

along the AI supply chain, wage and price pass-through in adopting firms, demand responsiveness in AI-exposed sectors, and the ownership of AI-related capital. Finally, for economy-wide propagation, the main gaps concern household saving responses and the composition of additional spending.

Even where relevant measures exist, the evidence often remains fragmented. AI providers observe how the technology is used but not its broader economic consequences, while statistical agencies observe economic outcomes but have limited visibility into AI use. A growing number of studies link AI adoption or exposure to labor-market outcomes, which is an important step, but establishing causal effects remains difficult. These analyses are also rarely linked to measures of market power, supply-chain bottlenecks, capital ownership, household income, or sectoral price and income elasticities of demand.

Building datasets that connect the indicators discussed in this section would provide a clearer picture of how AI's gains are distributed. Developing this infrastructure will be essential for determining which distributional path the economy is heading toward.

5 Conclusion

Artificial intelligence could raise living standards broadly, or it could concentrate its gains at the top while leaving much of the workforce behind. How those gains are distributed will have profound economic, social, and political consequences. This paper develops a unified framework for understanding the forces that will shape that distribution.

The framework distinguishes three components of inequality: the concentration of labor income (c^L_{top}), the concentration of capital income (c^K_{top}), and the division of income between labor and capital (s_K). It then traces how AI affects these components through three categories: its direct effects on production, the initial division of productivity gains, and the economy-wide demand effects generated as those gains are spent or saved. The central conclusion is that AI does not imply a single distributional outcome. The same technology can produce concentrated gains or shared prosperity depending on what AI can do, how firms deploy it, how markets divide the resulting gains, who owns the capital that benefits, and where subsequent demand is directed. Some of these forces are technological, but many depend on *policy* such as market structure, bargaining power,

access, and ownership.

The outcome is therefore not predetermined. Pre-redistribution policies that shape factors such as competition, the direction of technological development, workers' bargaining power and mobility, access to AI and relevant skills, and ownership of AI-related capital can influence how gains are distributed before taxes and transfers. Redistribution can then offset some of the concentration that market outcomes still produce. Just as importantly, many of the mechanisms identified in this paper can be observed before their effects appear in aggregate inequality statistics. Monitoring AI use, labor-market outcomes, market power, the division of productivity gains, and the economy-wide demand effects can provide early evidence of which path is emerging.

The distribution of AI's gains will be determined not by the technology alone, but by the markets, institutions, and policies surrounding it. Understanding the mechanisms behind AI's distributional effects, and the forces that shape them, is the first step toward building an economy in which AI expands prosperity and its gains are broadly shared.

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A Appendix

A.1 Pre-Tax versus Post-Tax Income Inequality

This paper studies inequality in *pre-tax* income. The primary reason is conceptual: technology affects the income distribution through markets—wages, profits, and prices—so its direct effects are reflected primarily in the pre-tax distribution. Taxes and transfers affect how much of those effects ultimately reach households, but they are largely a policy response to technological change rather than part of its direct impact. Informing that policy response is one of this paper's objectives. This appendix establishes a second, empirical reason for focusing on pre-tax income: across countries, and over time within countries, post-redistribution income inequality closely tracks pre-redistribution inequality. In other words, redistribution lowers the level of inequality, but historically most of any change in pre-tax inequality has carried over to post-tax inequality. Within countries, more than 80 percent of a change in pre-tax inequality shows up after taxes and transfers.

Data: We use the World Inequality Database (World Inequality Database, 2026), which builds on the distributional national accounts methodology (Piketty et al., 2018). For each economy and year, we take the shares of national income received by the top 1% and the bottom 50% under two income concepts: *pre-tax national income*, which includes social-insurance benefits net of contributions but excludes most other redistribution, and *post-tax national income*, measured after all taxes and transfers, including in-kind transfers. The sample covers 38 of the 43 economies in the IMF's April 2026 advanced-economy classification, excluding five whose WID series are entirely imputed from regional models.[26] For all 38 economies, WID's post-tax series come from Fisher-Post and Gethin (2023), who estimate them for 1980–2019; for later years, WID projects post-tax incomes by applying each economy's 2019 ratio of post-tax to pre-tax income at each point of the distribution. We therefore end the sample in 2019. The cross-sectional analysis uses 2019, and the panel specifications use 1,587 country-years: 1980–2019 for 37 economies and 1913–2019 for the United States, the only economy with post-tax estimates before 1980.

Method: We regress each post-tax income share on the corresponding pre-tax share.

[26] Andorra, Liechtenstein, Macao SAR, Puerto Rico, and San Marino, whose WID series all carry a data-quality score of zero. Including them changes the results very little. Shares refer to adults aged 20 and over, with income split equally between spouses. WID ranks individuals separately under each income concept, so the pre-tax and post-tax top 1% need not contain the same people.

We move from comparisons across countries to comparisons within them: a 2019 cross-section, a pooled panel, a panel with country fixed effects, and a panel with both country and year fixed effects. The country fixed-effects specification is our main specification. It compares each country with itself over time and asks whether, when pre-tax inequality rises, post-tax inequality rises by less; that is, whether redistribution increases. Two statistics are of primary interest. The R^2 measures how much of the variation in post-tax inequality is explained by pre-tax inequality. The slope measures *pass-through*: by how many percentage points the post-tax share changes when the pre-tax share changes by one point. A slope of one means that differences or changes in pre-tax inequality carry over fully to post-tax inequality; a slope below one means that redistribution partly offsets them. In both figures, the dashed 45-degree line marks one-for-one pass-through; in Figure A.1, the vertical distance between each economy and this line is the level effect of redistribution.

Results: Figure A.1 presents the cross-country evidence for 2019. Redistribution reduces the level of inequality in every economy: the post-tax top-1% share lies below the corresponding pre-tax share in all 38 economies, while the post-tax bottom-50% share lies above it. Yet pre-tax inequality remains a strong predictor of post-tax inequality. Across countries, it accounts for 82–87% of the variation in post-tax income shares, and differences in top income shares pass through one-for-one (slope = 0.99). For the bottom 50%, the slope exceeds one (1.47), suggesting that countries where the bottom half receives a larger pre-tax share also redistribute more toward it, amplifying rather than offsetting cross-country differences.

Figure A.2 presents our main, within-country evidence. A one-point rise in a country's pre-tax top-1% share is associated with a 0.83-point rise in its post-tax share, and a one-point fall in its pre-tax bottom-50% share with a 0.89-point fall in its post-tax share, with within-R^2 values of 0.90 and 0.76, respectively. Redistribution therefore offsets only a small part of within-country changes in pre-tax inequality.

Interpretation: Since 1980 in advanced economies—and over the past century in the United States—taxes and transfers have reduced the level of inequality but offset only a small part of differences and changes in it: within countries, most of a rise in pre-tax inequality has carried over to post-tax inequality. These are descriptive patterns rather than causal estimates. Still, they suggest that, unless redistribution responds much more

strongly than it has in the past, how AI shapes the pre-tax distribution is likely to be the main determinant of its effect on the distribution of household post-tax income. This motivates the paper's focus on pre-tax inequality.

Table A.1: Post-Tax on Pre-Tax Income-Share Regressions, IMF Advanced Economies

	Top 1% share		Bottom 50% share		
Specification	Slope	R^2	Slope	R^2	Obs.
Cross-section (2019)	0.989 (0.054)	0.869	1.465 (0.089)	0.818	38
Pooled panel	0.862 (0.030)	0.886	1.216 (0.087)	0.742	1,587
Country fixed effects	0.834 (0.033)	0.896	0.888 (0.043)	0.763	1,587
Country and year fixed effects	0.897 (0.040)	0.867	1.073 (0.045)	0.797	1,587

Notes: Each cell reports the slope from a regression of the post-tax income share on the corresponding pre-tax income share. Standard errors in parentheses: heteroskedasticity-robust in the cross-section and clustered by economy in the panel specifications. In the fixed-effects rows, R^2 is the within R^2. The panel covers 1980–2019 for 37 economies and 1913–2019 for the United States. Source: World Inequality Database.

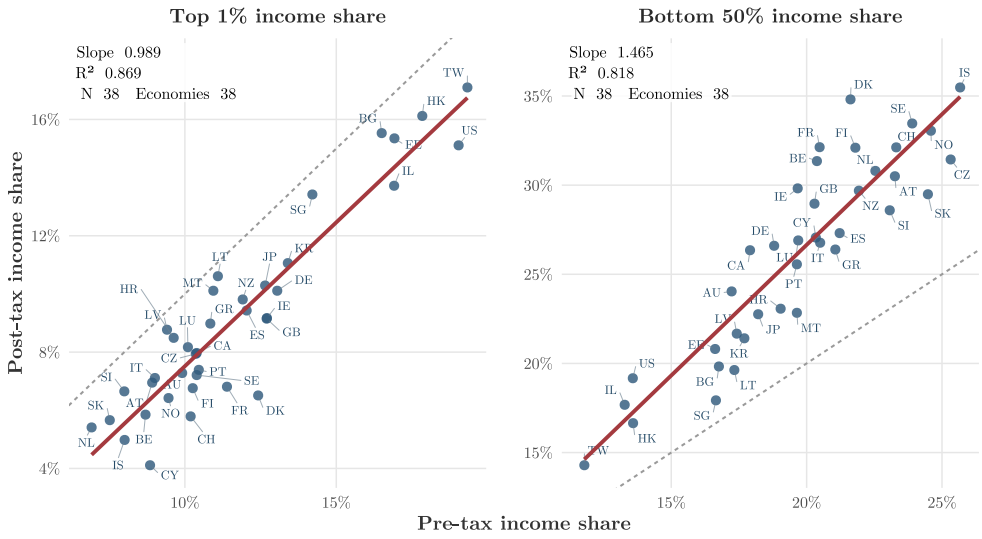


Figure A.1: Pre-Tax versus Post-Tax Income Shares Across Advanced Economies, 2019

Notes: Each point is one of 38 IMF advanced economies in 2019. The solid line is the fitted regression line. The dashed 45-degree line marks one-for-one pass-through; points below it (top 1%) or above it (bottom 50%) indicate that redistribution reduces inequality in levels. Source: World Inequality Database.



Figure A.2: Pre-Tax versus Post-Tax Income Shares Within Countries over Time

Notes: Both income shares are residualized on country fixed effects, so each point is a country-year deviation from the country's own average (1,587 country-years, 38 economies; 1980–2019, and 1913–2019 for the United States). By the Frisch–Waugh–Lovell theorem, the slope of the fitted line equals the country fixed-effects estimate in Table A.1; the reported R^2 is the within R^2. The dashed 45-degree line marks one-for-one pass-through. Source: World Inequality Database.

A.2 Derivation of the Top-Income-Share Decomposition

Let Y denote total income, Y_{top} the income received by the top group, and L_{top} and K_{top} that group's labor and capital income. Then:

$$\begin{aligned} S_{\text{top}} \equiv \frac{Y_{\text{top}}}{Y} &= \frac{L_{\text{top}} + K_{\text{top}}}{Y} \\ &= \frac{\frac{L_{\text{top}}}{L_{\text{total}}} L_{\text{total}} + \frac{K_{\text{top}}}{K_{\text{total}}} K_{\text{total}}}{Y} \\ &= c^L_{\text{top}} \frac{L_{\text{total}}}{Y} + c^K_{\text{top}} \frac{K_{\text{total}}}{Y} \\ &= (1 - s_K) c^L_{\text{top}} + s_K c^K_{\text{top}}. \end{aligned}$$

Here, $c^L_{\text{top}} = L_{\text{top}}/L_{\text{total}}$, $c^K_{\text{top}} = K_{\text{top}}/K_{\text{total}}$, and $s_K = K_{\text{total}}/Y$.

A.3 An Analogous Decomposition for the Gini Coefficient

The decomposition in the main text is exact because the top income share is linear in incomes (it is a share of a sum). The most familiar inequality measure, the Gini, admits an analogous decomposition by income source, at the cost of one extra term. Writing total income as labor plus capital income, the Gini of total income decomposes as

$$G \;=\; \underbrace{(1-s_K)}_{\text{labor share}}\, G_L\, R_L \;+\; \underbrace{s_K}_{\text{capital share}}\, G_K\, R_K,$$

where G_L, G_K are the Ginis of labor and of capital income (each source's own concentration) and R_L, R_K are the *Gini correlations* between each source and total income (how each source's ranking lines up with the ranking of total income).[27] This mirrors the three factors of the main text: each source enters through its **income share** (s_K, $1-s_K$) and its **own concentration** (G_K, G_L). The difference is the correlation term R_k: with the Gini the labor–capital overlap appears as an explicit factor, whereas in the top-share formula it is absorbed by defining the top group by total income. The economic logic is the same—a rising capital share raises overall inequality more when capital income is both unequally distributed (G_K high) and concentrated among the rich (R_K high)—but the expression is no longer a simple weighted average.

A.4 Data and construction of Figure 2

Figure 2 draws on task-level automatability ratings for U.S. occupations from Hosseini and Lichtinger (2026a), which build on the task-based exposure framework of Eloundou et al. (2024). Eloundou et al. rate each task in the O*NET database for its exposure to large language models and, in an exploratory rubric, for its automation potential on a five-level scale—from tasks an AI system cannot reliably perform at all, to tasks it could complete on its own without human oversight. Hosseini and Lichtinger re-rate tasks for automatability with a more capable model, assigning each of roughly 19,000 tasks across 923 occupations an automation tier from T0 (least automatable) to T4 (most). These ratings describe what AI could *potentially* automate, not what has been automated: they are

[27] $R_k = \mathrm{Cov}\big(y_k, F(y)\big) / \mathrm{Cov}\big(y_k, F(y_k)\big)$, where $F(y)$ is the rank by total income and $F(y_k)$ the rank by source k; $R_k \in [-1, 1]$.

produced by language models applying a rubric, and the automation rubric in particular is exploratory rather than validated against realized outcomes.

The figure treats this automation tier as a threshold and reports three scenarios: *limited* counts only fully-automatable tasks (T4); *modest* counts tasks rated clearly automatable (T3–T4, the binary measure used in the main text); and *broad* counts any task with at least partial automation potential (T2–T4). For each occupation we compute the share of its tasks automatable under each threshold, weighting tasks by their share of the occupation's labor time. We then attach each occupation's median annual wage and employment from the BLS Occupational Employment and Wage Statistics (2024), rank occupations by wage, and group them into employment-weighted deciles, each holding about a tenth of U.S. workers. The figure plots the employment-weighted average automatable share within each decile, separately for the three scenarios (864 occupations, about 142 million workers; because employment is reported at a broader occupational level, it is split evenly across the detailed occupations within each group to avoid double-counting). The underlying data and code are available from the authors upon request.